Tarea 3 Antonio Bustos

July 5, 2025

```
               sexo          edad          imce    vive_padre    vive_madre  \
count  31055.000000  31055.000000  31055.000000  31055.000000  31055.000000
mean       0.538979     82.589084      1.017787      0.720335      0.966511
std        0.498486      4.360270      1.370552      0.449415      0.189671
min        0.000000     62.000000     -5.020000      0.000000      0.000000
25%        0.000000     80.000000      0.120000      0.000000      1.000000
50%        1.000000     81.000000      0.970000      1.000000      1.000000
75%        1.000000     82.000000      1.930000      1.000000      1.000000
max        1.000000    107.000000      5.040000      2.000000      2.000000

                sk1           sk2           sk3           sk4           sk5  \
count  31055.000000  31055.000000  31055.000000  31055.000000  31055.000000
mean       1.114088      1.391177      1.262856      1.256352      1.268588
std        0.394004      0.650190      0.585766      0.578266      0.564929
min        1.000000      1.000000      1.000000      1.000000      1.000000
25%        1.000000      1.000000      1.000000      1.000000      1.000000
50%        1.000000      1.000000      1.000000      1.000000      1.000000
75%        1.000000      2.000000      1.000000      1.000000      1.000000
max        5.000000      5.000000      5.000000      5.000000      5.000000

       …           sk9          sk10          sk11          sk12  \
count  …  31055.000000  31055.000000  31055.000000  31055.000000
mean   …      1.333666      1.853743      1.385349      1.502785
std    …      0.666009      0.941171      0.668783      0.799375
min    …      1.000000      1.000000      1.000000      1.000000
25%    …      1.000000      1.000000      1.000000      1.000000
50%    …      1.000000      2.000000      1.000000      1.000000
75%    …      1.000000      2.000000      2.000000      2.000000
max    …      5.000000      5.000000      5.000000      5.000000

                sk13    act_fisica          area         educm         educp  \
count  31055.000000  30051.000000  31055.000000  30682.000000  31055.000000
mean       1.694735      2.556620      0.911544     13.069291     12.955756
std        0.984946      1.068041      0.283961      3.338320      3.425190
min        1.000000      1.000000      0.000000      0.000000      0.000000
25%        1.000000      2.000000      1.000000     12.000000     11.000000
50%        1.000000      2.000000      1.000000     13.000000     13.000000
```

```
75%        2.000000      3.000000      1.000000     15.000000     14.000000
max        5.000000      5.000000      1.000000     22.000000     22.000000

         madre_work
count  31055.000000
mean       0.092803
std        0.943509
min       -1.000000
25%       -1.000000
50%        0.000000
75%        1.000000
max        1.000000

[8 rows x 23 columns]
```

0.1 1. Limpieza de Datos

```
   index  sexo  edad  imce  vive_padre  vive_madre  sk1  sk2  sk3  sk4  …  \
0      1     0    76  0.71           0           1    1    1    1    1  …
1      3     1    84  2.05           1           1    1    1    1    1  …
2      4     0    86  1.05           1           1    1    1    1    1  …
3      5     0    74  1.39           1           1    1    2    1    1  …
4      6     1    91  2.75           1           1    1    1    1    2  …

   sk9  sk10  sk11  sk12  sk13  act_fisica  area  educm  educp  madre_work
0    1     1     1     1     1         5.0     0    8.0      8           1
1    1     1     1     1     1         2.0     1   16.0     12          -1
2    1     1     1     1     1         1.0     1   17.0     15           0
3    1     1     1     1     1         4.0     0    8.0      8          -1
4    3     3     3     2     2         2.0     1   20.0     19           1

[5 rows x 24 columns]

              index          sexo          edad          imce    vive_padre  \
count  29635.000000  29635.000000  29635.000000  29635.000000  29635.000000
mean   15558.514088      0.539194     82.545470      1.017501      0.719791
std     8937.216072      0.498470      4.296714      1.369436      0.449109
min        1.000000      0.000000     62.000000     -5.020000      0.000000
25%     7830.500000      0.000000     80.000000      0.120000      0.000000
50%    15556.000000      1.000000     81.000000      0.970000      1.000000
75%    23288.500000      1.000000     82.000000      1.930000      1.000000
max    31054.000000      1.000000    107.000000      5.040000      1.000000

         vive_madre           sk1           sk2           sk3           sk4  \
count  29635.000000  29635.000000  29635.000000  29635.000000  29635.000000
mean       0.974355      1.108419      1.386570      1.256757      1.250852
std        0.158078      0.379089      0.643402      0.575516      0.569380
```

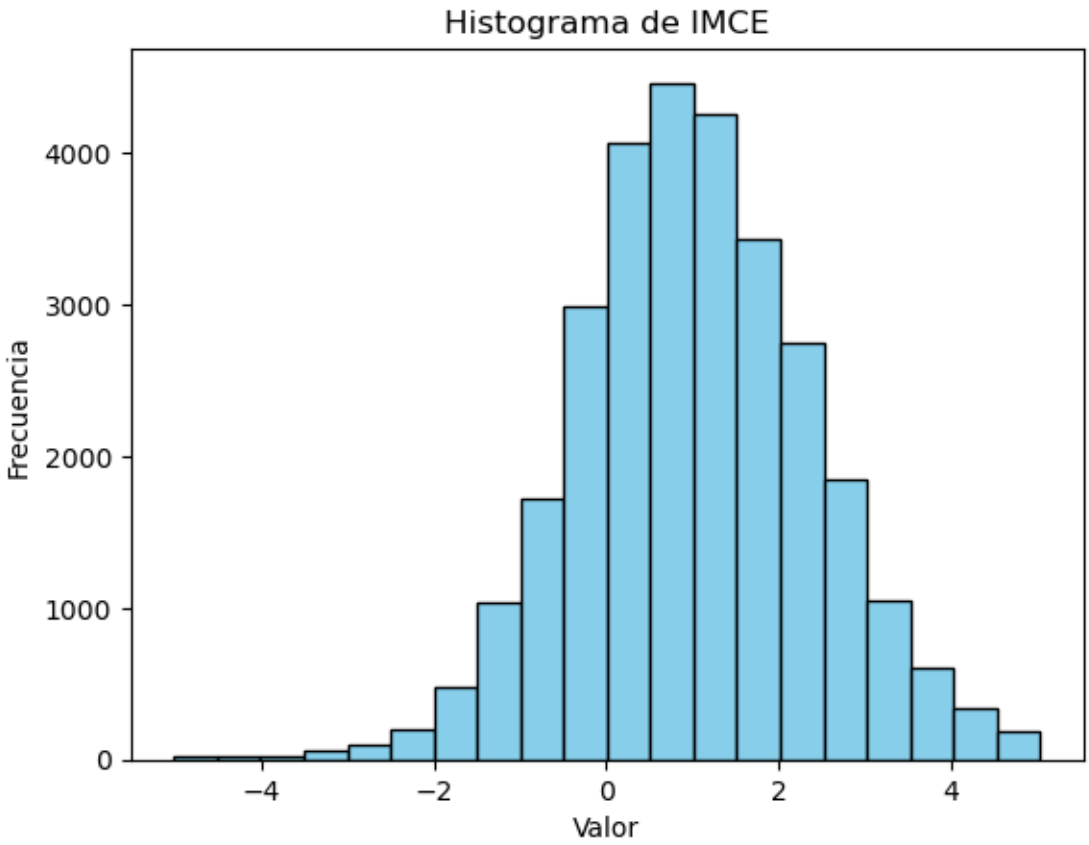
min	0.000000	1.000000	1.000000	1.000000	1.000000
25%	1.000000	1.000000	1.000000	1.000000	1.000000
50%	1.000000	1.000000	1.000000	1.000000	1.000000
75%	1.000000	1.000000	2.000000	1.000000	1.000000
max	1.000000	5.000000	5.000000	5.000000	5.000000

	…	sk9	sk10	sk11	sk12 \
count	…	29635.000000	29635.000000	29635.000000	29635.000000
mean	…	1.326303	1.844609	1.378843	1.495596
std	…	0.655024	0.933133	0.659927	0.790418
min	…	1.000000	1.000000	1.000000	1.000000
25%	…	1.000000	1.000000	1.000000	1.000000
50%	…	1.000000	2.000000	1.000000	1.000000
75%	…	1.000000	2.000000	2.000000	2.000000
max	…	5.000000	5.000000	5.000000	5.000000

	sk13	act_fisica	area	educm	educp \
count	29635.000000	29635.000000	29635.000000	29635.000000	29635.000000
mean	1.687025	2.556740	0.911625	13.106428	13.002396
std	0.979340	1.068159	0.283845	3.328033	3.417926
min	1.000000	1.000000	0.000000	0.000000	0.000000
25%	1.000000	2.000000	1.000000	12.000000	11.000000
50%	1.000000	2.000000	1.000000	13.000000	13.000000
75%	2.000000	3.000000	1.000000	15.000000	15.000000
max	5.000000	5.000000	1.000000	22.000000	22.000000

	madre_work
count	29635.000000
mean	0.107036
std	0.941576
min	-1.000000
25%	-1.000000
50%	1.000000
75%	1.000000
max	1.000000

[8 rows x 24 columns]

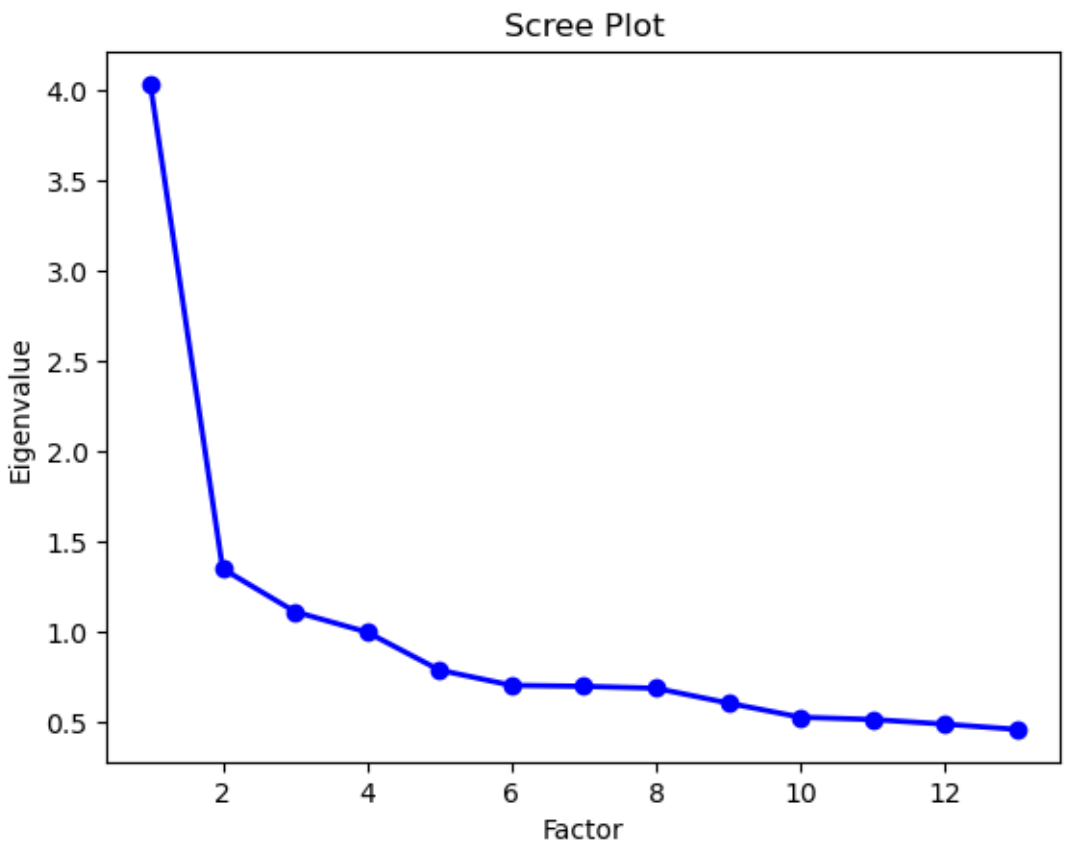


R: La data estaba bien, con pocos outliers o Nan por lo que solo se borraron esos Nan y se hizo un análisis a través de DataWrangler

0.2 2. EFA

```
array([[0.170956, 0.541000, 0.150807],
       [0.173792, 0.512760, 0.354520],
       [0.179809, 0.587227, 0.162152],
       [0.195005, 0.685408, 0.194815],
       [0.121379, 0.214461, 0.716558],
       [0.163073, 0.203981, 0.486387],
       [0.070645, 0.074728, 0.146490],
       [0.269462, 0.097361, 0.478533],
       [0.496403, 0.205356, 0.201783],
       [0.555717, 0.120762, 0.127795],
       [0.665475, 0.206294, 0.202347],
       [0.524699, 0.109824, 0.141209],
       [0.480038, 0.150919, 0.132026]])
```

```
(array([4.034214, 1.352109, 1.114244, 1.000843, 0.791244, 0.706343,
        0.701189, 0.690190, 0.608368, 0.529874, 0.517230, 0.491686,
        0.462466]),
 array([3.438924, 0.747141, 0.522044, 0.216095, 0.117614, 0.062805,
        0.007217, -0.011140, -0.034171, -0.063444, -0.081942, -0.084678,
        -0.128355]))

array([0.344650, 0.418811, 0.403460, 0.545763, 0.574182, 0.304773,
       0.032034, 0.311083, 0.329303, 0.339736, 0.526358, 0.307311,
       0.270644])
```




```
(array([1.752553, 1.607039, 1.348517]),
 array([0.134812, 0.123618, 0.103732]),
 array([0.134812, 0.258430, 0.362162]))
```

```
eta1 =~ sk11 + sk9 + sk10 + sk12 + sk13 + sk4
eta2 =~ sk7 + sk6
eta3 =~ sk4 + sk11 + sk2 + sk5 + sk3 + sk9 + sk1 + sk8 + sk6 + sk13
```

R: De este análisis, se puede observar claramente que la variable ‘SK7’ no es parte de ningún factor

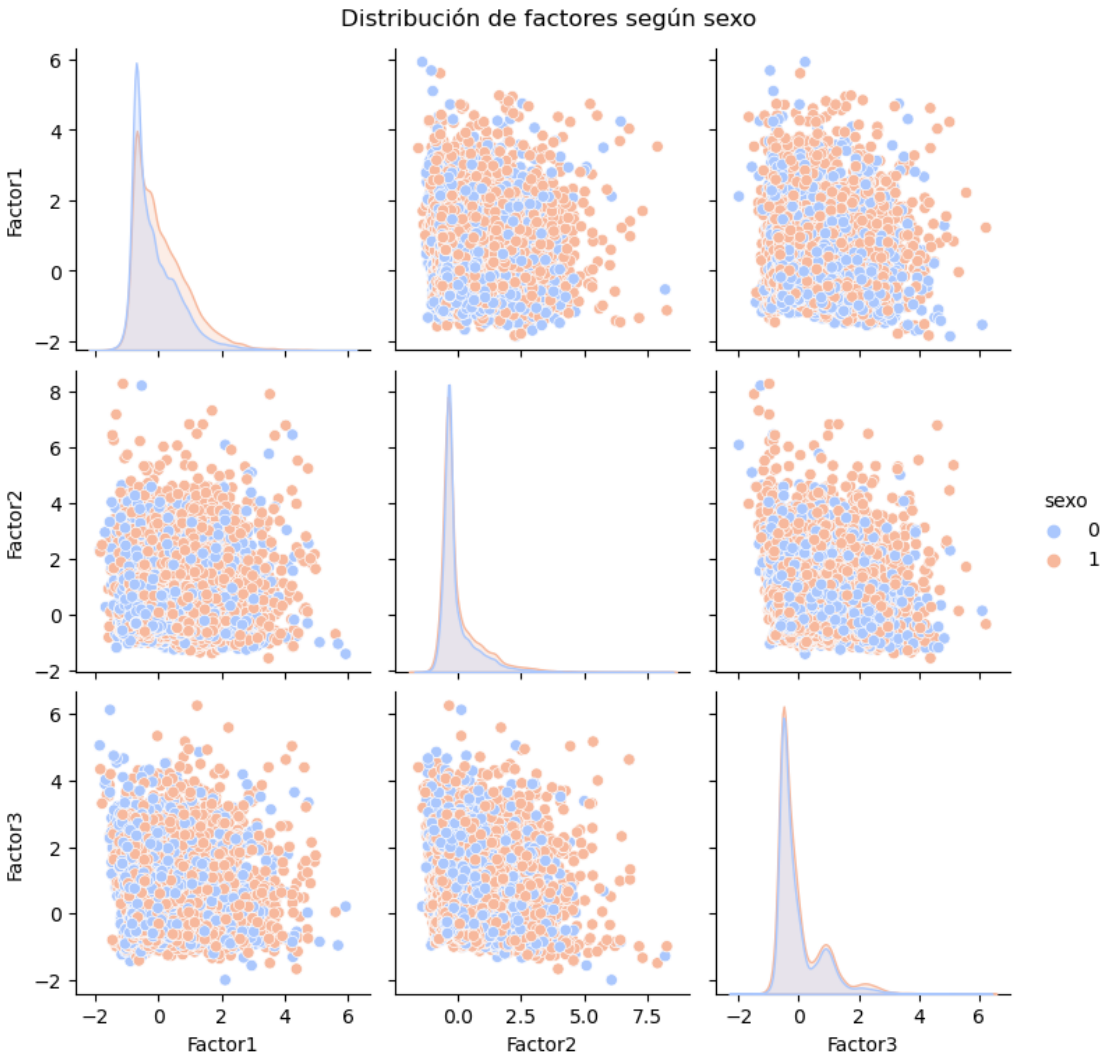
y no es explicado bien por ningún factor como se muestra en las comunalidades (3%), por lo que no se considerará para el modelo confirmatorio. Además, se puede ver por el método del codo que 3 Factores es lo correcto en este análisis, agregando que sus EigenValues son mayores que 1.

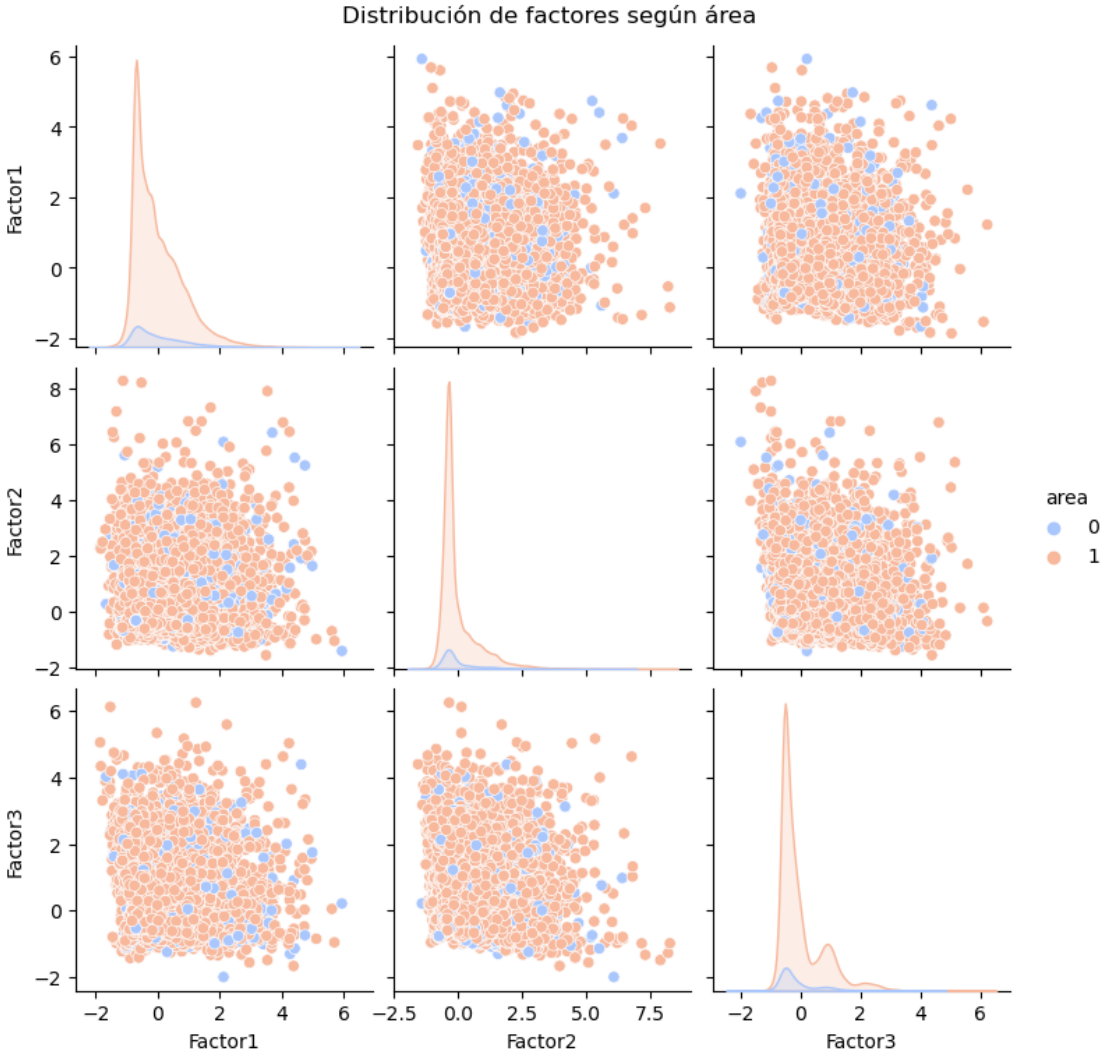
Por otro lado, el modelo exploratorio que entrega Semopy no es el mejor, ya que considera que las variables pueden estar en más de un factor y no muestra buenos factores (a mi criterio).

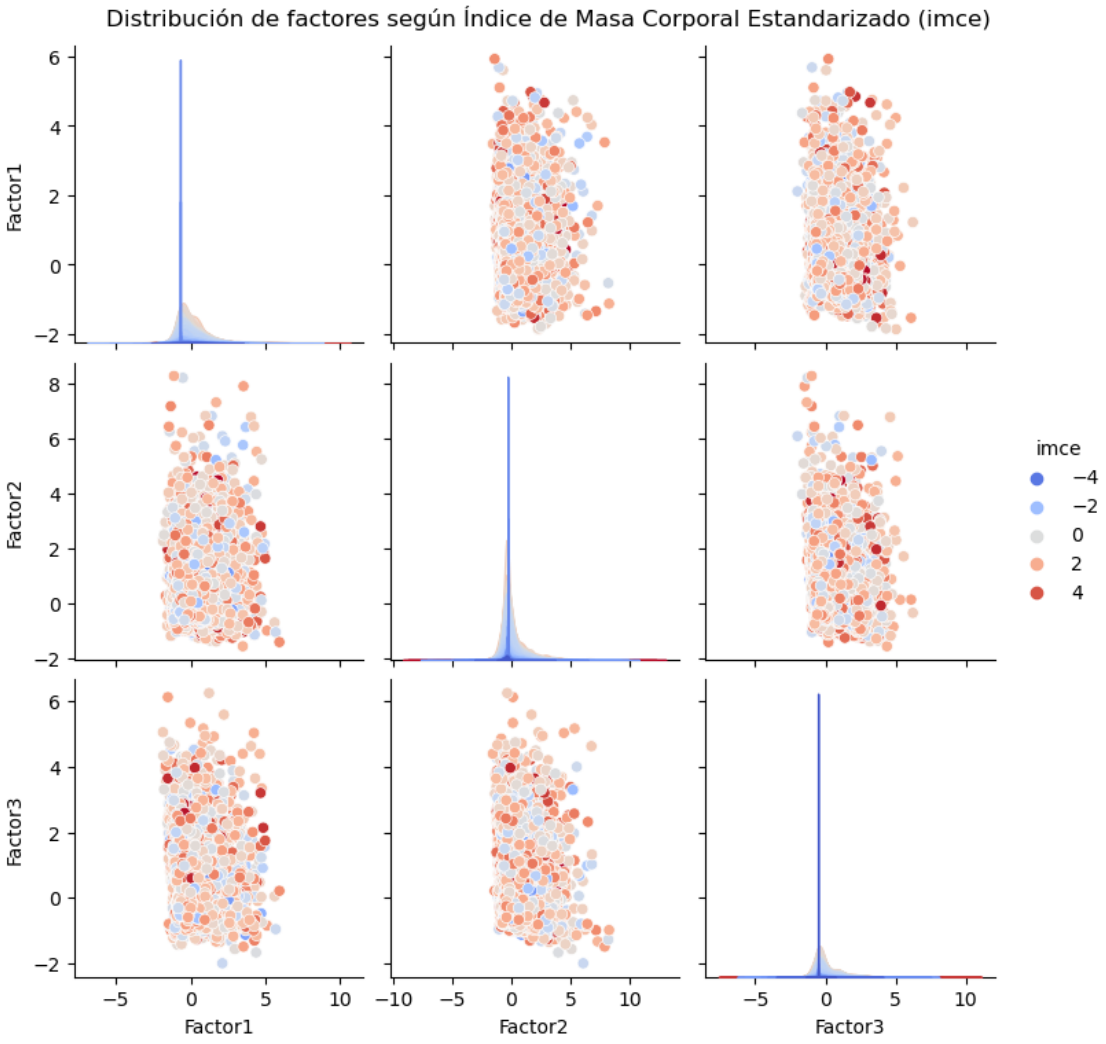
Además, podemos agregar que la varianza explicada de las variables en los factores ronda entre los 30%-58% (Según las comunalidades).

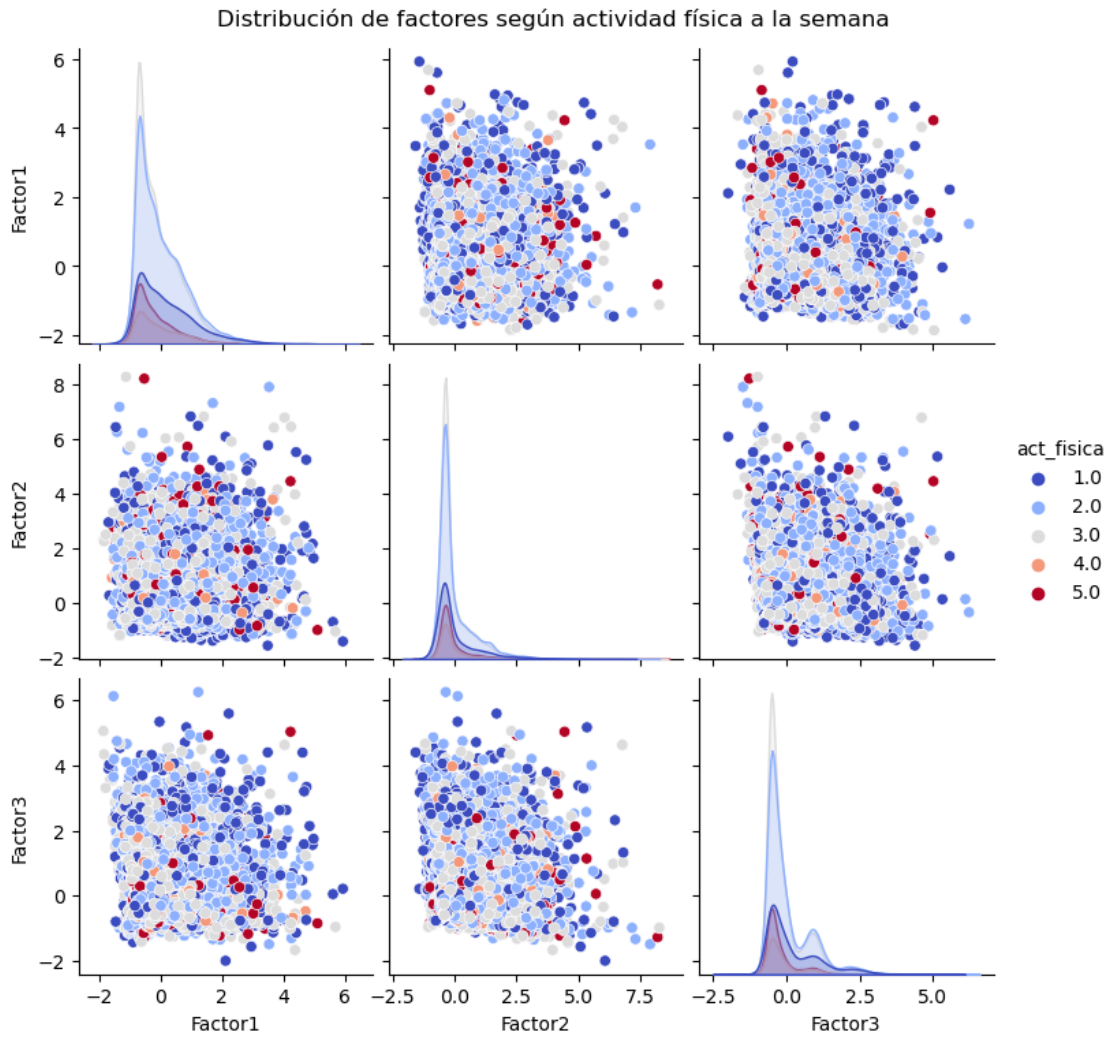
0.3 3. Diferencias del Modelo con Variables

	Factor1	Factor2	Factor3	sexo	area	imce	act_fisica
0	-0.711178	-0.311964	-0.400967	0	0	0.71	5.0
1	-0.708049	-0.316938	-0.484848	1	1	2.05	2.0
2	-0.709614	-0.314451	-0.442907	0	1	1.05	1.0
3	-0.787520	0.017855	-0.246423	0	0	1.39	4.0
4	1.759600	-0.071503	0.817094	1	1	2.75	2.0




Distribución de factores según área
Factor1
Factor2
Factor3
area
0
1


Distribución de factores según Índice de Masa Corporal Estandarizado (imce)
Factor1
Factor2
Factor3
imce
−4
−2
0
2
4



R: Según lo visto en los gráficos, podemos decir que no existe una separación clara de las variables en los factores, por lo que se ven dispersas los valores de las variables en todos los factores. Lo anterior, puede deberse a que las variables analizadas no fueron parte del análisis factorial, por lo que el Análisis no considero ningún tipo de separación con estas variables.

0.4 4. CFA

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.143
Number of iterations: 41
Params: 1.347 1.246 1.098 1.272 1.876 1.645 1.844 1.021 1.159 0.049 0.152 0.051
0.091 0.160 0.057 0.095 0.596 0.200 0.442 0.714 0.242 0.199 0.158 0.151 0.372
0.470 0.277
```

	lval	op	rval	Estimate	Est. Std	Std. Err	z-value	\
0	sk9	~	Curiosidad	1.000000	0.594873	-	-	
1	sk10	~	Curiosidad	1.346839	0.562305	0.01854	72.644149	
2	sk11	~	Curiosidad	1.245772	0.735504	0.014724	84.606271	
3	sk12	~	Curiosidad	1.097571	0.541108	0.015529	70.676835	
4	sk13	~	Curiosidad	1.271792	0.505956	0.018913	67.243421	
5	sk1	~	Afectividad	1.000000	0.583084	-	-	
6	sk2	~	Afectividad	1.876071	0.644674	0.023794	78.846648	
7	sk3	~	Afectividad	1.645058	0.631947	0.021115	77.909473	
8	sk4	~	Afectividad	1.843514	0.715743	0.022152	83.220784	
9	sk5	~	Sociabilidad	1.000000	0.717647	-	-	
10	sk6	~	Sociabilidad	1.021421	0.556959	0.014356	71.147471	
11	sk8	~	Sociabilidad	1.158700	0.560768	0.016214	71.462021	
12	Afectividad	~~	Afectividad	0.048866	1.000000	0.001008	48.484878	
13	Curiosidad	~~	Curiosidad	0.151799	1.000000	0.003074	49.376039	
14	Curiosidad	~~	Afectividad	0.051233	0.594859	0.000965	53.103886	
15	Curiosidad	~~	Sociabilidad	0.091166	0.584176	0.001671	54.573071	
16	Sociabilidad	~~	Sociabilidad	0.160438	1.000000	0.002865	56.008431	
17	Sociabilidad	~~	Afectividad	0.057348	0.647676	0.000997	57.5253	
18	sk1	~~	sk1	0.094864	0.660013	0.000914	103.735199	
19	sk10	~~	sk10	0.595514	0.683813	0.005669	105.044681	
20	sk11	~~	sk11	0.199903	0.459034	0.002529	79.03953	
21	sk12	~~	sk12	0.441682	0.707203	0.004134	106.829763	
22	sk13	~~	sk13	0.713596	0.744009	0.006523	109.394378	
23	sk2	~~	sk2	0.241844	0.584395	0.002502	96.677526	
24	sk3	~~	sk3	0.198897	0.600643	0.002022	98.350929	
25	sk4	~~	sk4	0.158107	0.487712	0.001868	84.62758	
26	sk5	~~	sk5	0.151081	0.484983	0.002176	69.415088	
27	sk6	~~	sk6	0.372212	0.689796	0.003731	99.754569	
28	sk8	~~	sk8	0.469584	0.685539	0.004729	99.288337	
29	sk9	~~	sk9	0.277164	0.646126	0.002721	101.873612	

	p-value
0	-
1	0.0
2	0.0
3	0.0
4	0.0
5	-
6	0.0
7	0.0
8	0.0
9	-
10	0.0
11	0.0
12	0.0
13	0.0

```
14     0.0
15     0.0
16     0.0
17     0.0
18     0.0
19     0.0
20     0.0
21     0.0
22     0.0
23     0.0
24     0.0
25     0.0
26     0.0
27     0.0
28     0.0
29     0.0

       DoF  DoF Baseline         chi2  chi2 p-value  chi2 Baseline       CFI  \
Value   51            66  4230.150078           0.0   80730.577968  0.948191

            GFI     AGFI       NFI       TLI     RMSEA        AIC         BIC  \
Value  0.947602  0.93219  0.947602  0.932953  0.052585  53.714517  277.725724

         LogLik
Value  0.142742
```

R: Según la relación y la carga que tienen las distintas variables en los factores, se hizo el modelo confirmatorio, obteniendo la siguiente estructura:

```
Curiosidad = sk9 + sk10 + sk11 + sk12 + sk13
Afectividad = sk1 + sk2 + sk3 + sk4
Sociabilidad = sk5 + sk6 + sk8
```

Con esto podemos ver que las estimaciones para cada variable en el factor son significativas (p-value<0.5) y que, el modelo se ajusta bien a los datos y los factores están bien definidos según los estadísticos,

```
TLI = 0.933
RMSEA = 0.053
CFI = 0.9482
```

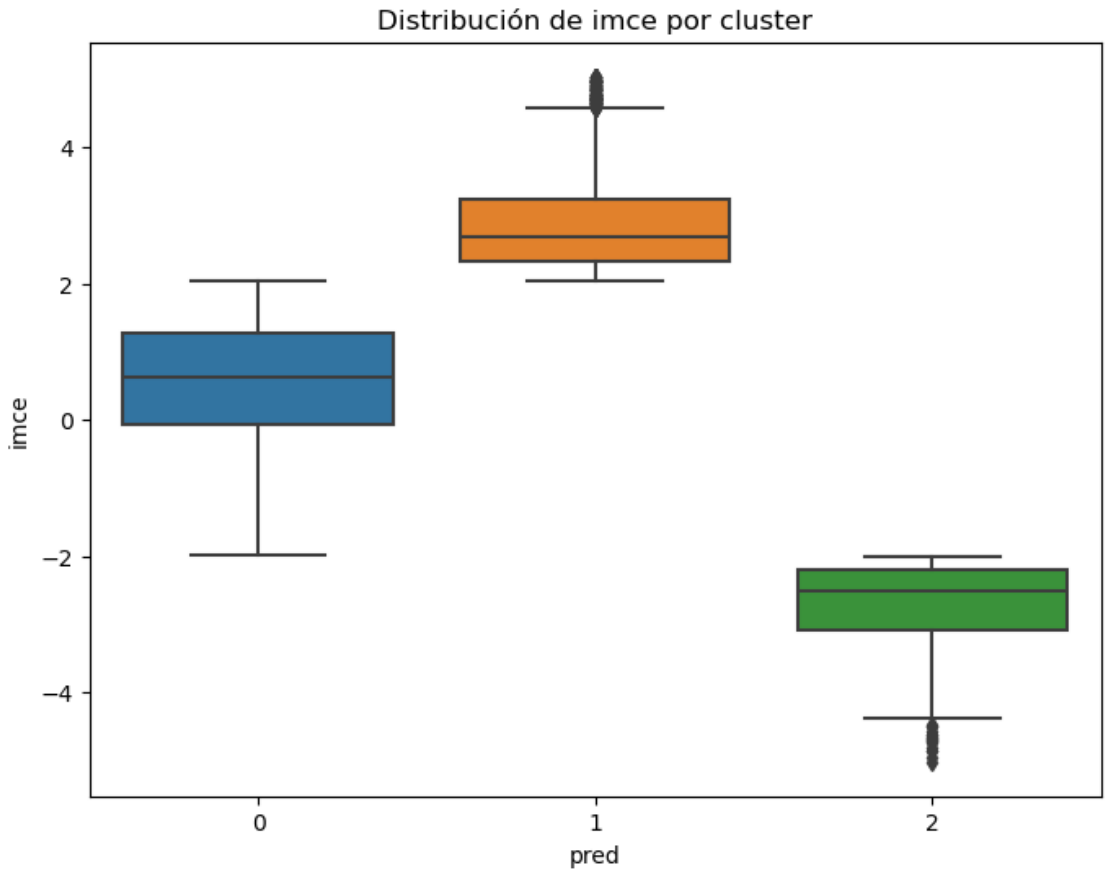
Lo que indica que las relaciones entre variables y factor parecen apropiadas.

0.5 5. Clase Latente

```
Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:00<00:00, 33.20it/s,
max_LL=-5.5e+4, max_avg_LL=-1.86]

Fitting StepMix…
```

```
Initializations (n_init) : 100%|      | 1/1 [00:01<00:00,  1.32s/it,
max_LL=-5.17e+4, max_avg_LL=-1.74]

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:03<00:00,  3.22s/it,
max_LL=-5.13e+4, max_avg_LL=-1.73]

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:12<00:00, 12.73s/it,
max_LL=-5.12e+4, max_avg_LL=-1.73]
/Users/antonio/anaconda3/envs/nuevo_entorno/lib/python3.11/site-
packages/stepmix/stepmix.py:968: ConvergenceWarning: Initializations did not
converge. Try different init parameters, or increase max_iter, abs_tol, rel_tol
or check for degenerate data.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:15<00:00, 15.45s/it,
max_LL=-5.12e+4, max_avg_LL=-1.73]
/Users/antonio/anaconda3/envs/nuevo_entorno/lib/python3.11/site-
packages/stepmix/stepmix.py:968: ConvergenceWarning: Initializations did not
converge. Try different init parameters, or increase max_iter, abs_tol, rel_tol
or check for degenerate data.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:15<00:00, 15.02s/it,
max_LL=-5.12e+4, max_avg_LL=-1.73]

   n_components            BIC            AIC  LogLikelihood  Relative Entropy
0             1  110050.091266  110041.794554      -1.856585               NaN
1             2  103429.704867  103404.814733      -1.744539          0.430243
2             3  102567.089024  102525.605467      -1.729637          0.647764
3             4  102531.108601  102473.031622      -1.728683          0.459772
4             5  102549.480147  102474.809745      -1.728645          0.325472
5             6  102569.213255  102477.949430      -1.728631          0.255846


/Users/antonio/anaconda3/envs/nuevo_entorno/lib/python3.11/site-
packages/stepmix/stepmix.py:968: ConvergenceWarning: Initializations did not
converge. Try different init parameters, or increase max_iter, abs_tol, rel_tol
or check for degenerate data.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:03<00:00,  3.45s/it,
max_LL=-5.13e+4, max_avg_LL=-1.73]
```

```
==============================================================================
MODEL REPORT
==============================================================================
    ==========================================================================
    Measurement model parameters
    ==========================================================================
          model_name     gaussian_unit
          class_no                   0      1       2
          param variable
          means imce            0.5993  2.407 -1.9462
    ==========================================================================
    Class weights
    ==========================================================================
        Class 1 : 0.71
        Class 2 : 0.27
        Class 3 : 0.02
    ==========================================================================
    Fit for 3 latent classes
    ==========================================================================
    Estimation method             : 1-step
    Number of observations        : 29635
    Number of latent classes      : 3
    Number of estimated parameters: 5
    Log-likelihood (LL)           : -51257.8027
    -2LL                          : 102515.6055
    Average LL                    : -1.7296
    AIC                           : 102525.61
    BIC                           : 102567.09
    CAIC                          : 102572.09
    Sample-Size Adjusted BIC      : 102602.68
    Entropy                       : 11467.8729
    Scaled Relative Entropy       : 0.6478


count    29635.000000
mean         0.252674
std          0.465593
min          0.000000
25%          0.000000
50%          0.000000
75%          0.000000
max          2.000000
Name: pred, dtype: float64
```




```
Las medias de los clusters son,
pred
0    0.541010
1    2.865563
2   -2.745749
Name: imce, dtype: float64
```

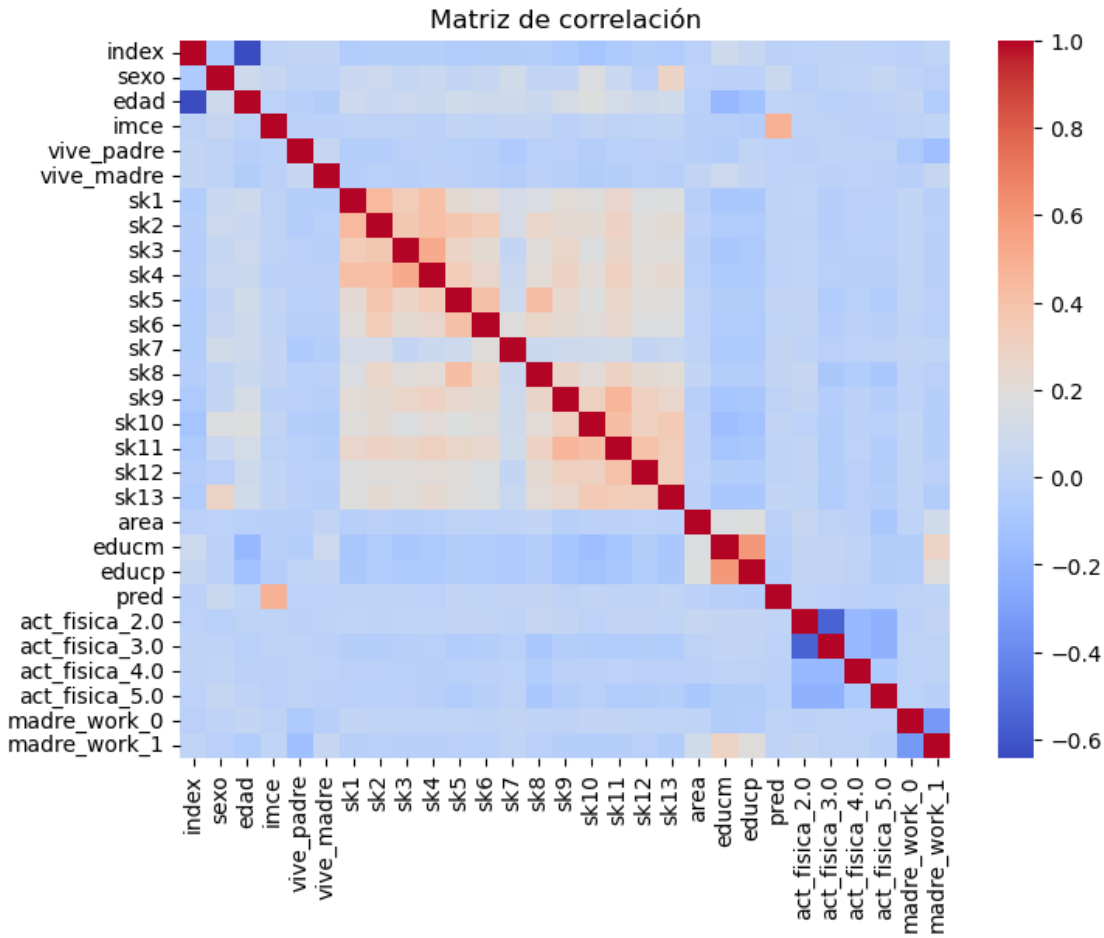
R: Para analizar cuantas clases son buenas, se hizo una comparación entre números de clases de 1 al 6, teniendo mejor resultado con 3 clases, donde el BIC y la Entropía Relativa tenían buenos valores en conjunto respecto a los otros escenarios.

Con esto, se aplicó el método de clases latentes, resultando 3 clases marcadas y definiéndolas como:

```
1. Clase 1 : Estándar o Promedio
2. Clase 2 : Sobrepeso
3. Clase 3 : Desnutrición
```

Además se hizo una predicción y se ve claramente en el gráfico de Caja (BoxPlot) como se distribuye el IMCE por cada cluster.

0.6 6. Modelo SEM




```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.353
Number of iterations: 53
Params: 1.346 1.245 1.098 1.271 1.878 1.646 1.845 1.022 1.160 -0.050 -0.319
0.246 -0.005 -0.014 0.136 0.008 0.002 -0.089 -0.116 0.051 0.088 -0.140 -0.085
-0.013 0.049 0.152 0.051 0.091 0.160 0.057 1.858 0.095 0.596 0.200 0.442 0.714
0.242 0.199 0.158 0.151 0.372 0.469 0.277
```

	lval	op	rval	Estimate	Est. Std	Std. Err	\
0	sk9	~	Curiosidad	1.000000	0.595084	-	
1	sk10	~	Curiosidad	1.345665	0.562077	0.018525	
2	sk11	~	Curiosidad	1.245159	0.735459	0.014713	
3	sk12	~	Curiosidad	1.098075	0.541460	0.015526	

4	sk13	~	Curiosidad	1.270820	0.505779	0.0189
5	sk1	~	Afectividad	1.000000	0.582865	-
6	sk2	~	Afectividad	1.877535	0.644849	0.023812
7	sk3	~	Afectividad	1.645562	0.631885	0.021125
8	sk4	~	Afectividad	1.844906	0.715942	0.022169
9	sk5	~	Sociabilidad	1.000000	0.717456	-
10	sk6	~	Sociabilidad	1.021569	0.556818	0.014348
11	sk8	~	Sociabilidad	1.160494	0.561468	0.016211
12	imce	~	Curiosidad	-0.049758	-0.014161	0.035645
13	imce	~	Afectividad	-0.318734	-0.051426	0.070302
14	imce	~	Sociabilidad	0.246367	0.072037	0.040734
15	imce	~	educm	-0.005306	-0.012895	0.003057
16	imce	~	educp	-0.013900	-0.034690	0.002909
17	imce	~	sexo	0.136468	0.049669	0.015924
18	imce	~	act_fisica_2.0	0.007761	0.002704	0.024323
19	imce	~	act_fisica_3.0	0.002012	0.000706	0.024192
20	imce	~	act_fisica_4.0	-0.089198	-0.014584	0.039994
21	imce	~	act_fisica_5.0	-0.115924	-0.023004	0.034655
22	imce	~	madre_work_0	0.051209	0.011317	0.028145
23	imce	~	madre_work_1	0.087885	0.032085	0.01793
24	imce	~	area	-0.139906	-0.028996	0.028595
25	imce	~	vive_madre	-0.084911	-0.009801	0.050377
26	imce	~	vive_padre	-0.013124	-0.004303	0.018057
27	Afectividad	~~	Afectividad	0.048827	1.000000	0.001007
28	Curiosidad	~~	Curiosidad	0.151922	1.000000	0.003076
29	Curiosidad	~~	Afectividad	0.051232	0.594838	0.000965
30	Curiosidad	~~	Sociabilidad	0.091192	0.584253	0.001671
31	Sociabilidad	~~	Sociabilidad	0.160358	1.000000	0.002862
32	Sociabilidad	~~	Afectividad	0.057326	0.647850	0.000996
33	imce	~~	imce	1.858127	0.990659	0.01532
34	sk1	~~	sk1	0.094894	0.660268	0.000914
35	sk10	~~	sk10	0.595664	0.684069	0.005669
36	sk11	~~	sk11	0.199921	0.459099	0.002529
37	sk12	~~	sk12	0.441633	0.706821	0.004135
38	sk13	~~	sk13	0.713754	0.744188	0.006524
39	sk2	~~	sk2	0.241800	0.584170	0.002501
40	sk3	~~	sk3	0.198922	0.600722	0.002022
41	sk4	~~	sk4	0.158037	0.487427	0.001868
42	sk5	~~	sk5	0.151172	0.485256	0.002174
43	sk6	~~	sk6	0.372408	0.689953	0.003731
44	sk8	~~	sk8	0.469097	0.684754	0.004727
45	sk9	~~	sk9	0.277084	0.645875	0.00272

	z-value	p-value
0	-	-
1	72.640607	0.0
2	84.631766	0.0

3	70.726066	0.0
4	67.23933	0.0
5	-	-
6	78.849563	0.0
7	77.895373	0.0
8	83.221171	0.0
9	-	-
10	71.201597	0.0
11	71.586634	0.0
12	-1.395937	0.162734
13	-4.533778	0.000006
14	6.048161	0.0
15	-1.735729	0.082612
16	-4.778575	0.000002
17	8.56998	0.0
18	0.319102	0.749649
19	0.083157	0.933727
20	-2.230266	0.02573
21	-3.345065	0.000823
22	1.819451	0.068843
23	4.901467	0.000001
24	-4.89264	0.000001
25	-1.685486	0.091895
26	-0.726803	0.467347
27	48.468538	0.0
28	49.397326	0.0
29	53.105174	0.0
30	54.585348	0.0
31	56.035824	0.0
32	57.527649	0.0
33	121.28597	0.0
34	103.769061	0.0
35	105.066674	0.0
36	79.054341	0.0
37	106.803191	0.0
38	109.407393	0.0
39	96.671598	0.0
40	98.375567	0.0
41	84.611249	0.0
42	69.551589	0.0
43	99.816059	0.0
44	99.247535	0.0
45	101.853286	0.0

	DoF	DoF Baseline	chi2	chi2 p-value	chi2 Baseline	CFI	\
Value	282	312	10467.635443	0.0	135090.518247	0.924427	

```
            GFI      AGFI       NFI       TLI     RMSEA        AIC  \
Value  0.922514  0.914271  0.922514  0.916387  0.034912  85.293563

              BIC    LogLik
Value  442.052152  0.353219
```

R: Podemos ver que existen pocos valores estimados no significativos y que además, el modelo es aceptable considerando que,

```
TLI = 0.9164
RMSEA = 0.0349
CFI = 0.9244
```

Además, se intentó hacer modelos estructurales donde se buscaba encontrar relaciones entre factores y otras variables, para ver si los factores eran explicados por otras variables pero no se logró nada bueno por lo que se decidió dejarlo simplemente lineal y con un modelo estructural.

0.7 7. Modelo por cada Cluster

0.7.1 Cluster 1: Peso Estándar

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.353
Number of iterations: 45
Params: 1.347 1.252 1.089 1.248 1.891 1.692 1.899 1.034 1.174 -0.026 -0.063
0.021 0.002 -0.009 -0.023 0.006 0.010 -0.046 -0.013 0.017 0.040 -0.061 -0.092
-0.006 0.047 0.149 0.050 0.088 0.153 0.055 0.823 0.095 0.586 0.196 0.435 0.704
0.241 0.194 0.158 0.147 0.365 0.451 0.278

            lval  op            rval  Estimate  Est. Std  Std. Err  \
0            sk9   ~      Curiosidad  1.000000  0.590859         -
1           sk10   ~      Curiosidad  1.347160  0.562319  0.021355
2           sk11   ~      Curiosidad  1.251834  0.737497  0.017056
3           sk12   ~      Curiosidad  1.088565  0.537531  0.017817
4           sk13   ~      Curiosidad  1.248077  0.498218   0.02161
5            sk1   ~     Afectividad  1.000000  0.575790         -
6            sk2   ~     Afectividad  1.890978  0.640734  0.027799
7            sk3   ~     Afectividad  1.692356  0.639494  0.024907
8            sk4   ~     Afectividad  1.899105  0.719530  0.026313
9            sk5   ~    Sociabilidad  1.000000  0.714426         -
10           sk6   ~    Sociabilidad  1.033997  0.556272  0.016675
11           sk8   ~    Sociabilidad  1.173842  0.563965  0.018763
12          imce   ~      Curiosidad -0.025666 -0.010916  0.027565
13          imce   ~     Afectividad -0.063122 -0.015054  0.055103
14          imce   ~    Sociabilidad  0.021230  0.009138  0.031877
15          imce   ~           educm  0.001565  0.005734  0.002349
```

16	imce	~	educp	-0.009456	-0.035604	0.002234
17	imce	~	sexo	-0.022562	-0.012411	0.012112
18	imce	~	act_fisica_2.0	0.005856	0.003068	0.018638
19	imce	~	act_fisica_3.0	0.010428	0.005523	0.018497
20	imce	~	act_fisica_4.0	-0.045801	-0.011405	0.030289
21	imce	~	act_fisica_5.0	-0.012741	-0.003898	0.026052
22	imce	~	madre_work_0	0.017289	0.005742	0.021501
23	imce	~	madre_work_1	0.040093	0.022072	0.013657
24	imce	~	area	-0.060771	-0.018728	0.0221
25	imce	~	vive_madre	-0.091983	-0.015961	0.038512
26	imce	~	vive_padre	-0.005640	-0.002784	0.013788
27	Afectividad	~~	Afectividad	0.046918	1.000000	0.001124
28	Curiosidad	~~	Curiosidad	0.149205	1.000000	0.003493
29	Curiosidad	~~	Afectividad	0.050259	0.600693	0.001087
30	Curiosidad	~~	Sociabilidad	0.088480	0.585974	0.001864
31	Sociabilidad	~~	Sociabilidad	0.152809	1.000000	0.003139
32	Sociabilidad	~~	Afectividad	0.055204	0.651967	0.001103
33	imce	~~	imce	0.822624	0.997276	0.007747
34	sk1	~~	sk1	0.094600	0.668466	0.001037
35	sk10	~~	sk10	0.585577	0.683798	0.006393
36	sk11	~~	sk11	0.196071	0.456098	0.002866
37	sk12	~~	sk12	0.435102	0.711060	0.004658
38	sk13	~~	sk13	0.703911	0.751779	0.007343
39	sk2	~~	sk2	0.240887	0.589460	0.002834
40	sk3	~~	sk3	0.194211	0.591047	0.002281
41	sk4	~~	sk4	0.157630	0.482276	0.002148
42	sk5	~~	sk5	0.146580	0.489596	0.002391
43	sk6	~~	sk6	0.364599	0.690561	0.004186
44	sk8	~~	sk8	0.451453	0.681943	0.005232
45	sk9	~~	sk9	0.278177	0.650886	0.003119

	z-value	p-value
0	-	-
1	63.083365	0.0
2	73.396283	0.0
3	61.096157	0.0
4	57.754375	0.0
5	-	-
6	68.023684	0.0
7	67.946127	0.0
8	72.173458	0.0
9	-	-
10	62.007665	0.0
11	62.560517	0.0
12	-0.931093	0.351806
13	-1.145539	0.251986
14	0.665995	0.505414

```
15    0.666281  0.505231
16   -4.233358  0.000023
17   -1.862831  0.062486
18    0.314217  0.753357
19    0.563727   0.57294
20   -1.512147  0.130496
21    -0.48906  0.624799
22    0.804096  0.421342
23    2.935695  0.003328
24   -2.749756  0.005964
25   -2.388452   0.01692
26   -0.409036  0.682513
27   41.734389       0.0
28   42.717786       0.0
29   46.228968       0.0
30   47.468447       0.0
31   48.684997       0.0
32   50.037215       0.0
33  106.192662       0.0
34    91.23596       0.0
35   91.600617       0.0
36    68.40448       0.0
37   93.413382       0.0
38   95.857094       0.0
39   84.994653       0.0
40   85.137346       0.0
41     73.3976       0.0
42   61.314338       0.0
43   87.107383       0.0
44    86.28156       0.0
45    89.19204       0.0

       DoF  DoF Baseline         chi2  chi2 p-value  chi2 Baseline       CFI  \
Value  282           312  7955.284347           0.0  103257.874336  0.925463

            GFI      AGFI       NFI       TLI     RMSEA        AIC  \
Value  0.922957  0.914761  0.922957  0.917533  0.034729  85.294776

              BIC    LogLik
Value  430.325831  0.352612
```

0.7.2 Cluster 2: Sobrepeso

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.393
```

Number of iterations: 42
Params: 1.353 1.234 1.142 1.347 1.805 1.479 1.642 0.981 1.107 -0.036 -0.169 0.158 -0.008 -0.004 0.179 -0.032 0.002 -0.069 -0.030 -0.004 -0.017 0.008 0.073 -0.011 0.057 0.158 0.054 0.100 0.186 0.065 0.443 0.096 0.626 0.211 0.468 0.742 0.245 0.215 0.159 0.165 0.397 0.524 0.272

	lval	op	rval	Estimate	Est. Std	Std. Err	z-value	\
0	sk9	~	Curiosidad	1.000000	0.605464	-	-	
1	sk10	~	Curiosidad	1.352749	0.561573	0.038821	34.845788	
2	sk11	~	Curiosidad	1.234025	0.729829	0.030306	40.718775	
3	sk12	~	Curiosidad	1.142368	0.552518	0.033173	34.43696	
4	sk13	~	Curiosidad	1.347416	0.527563	0.040493	33.275416	
5	sk1	~	Afectividad	1.000000	0.611815	-	-	
6	sk2	~	Afectividad	1.804563	0.657697	0.046276	38.995369	
7	sk3	~	Afectividad	1.478587	0.607239	0.039907	37.050408	
8	sk4	~	Afectividad	1.642195	0.702171	0.040647	40.400966	
9	sk5	~	Sociabilidad	1.000000	0.727136	-	-	
10	sk6	~	Sociabilidad	0.981093	0.557132	0.02906	33.761372	
11	sk8	~	Sociabilidad	1.107213	0.550340	0.033059	33.491941	
12	imce	~	Curiosidad	-0.036303	-0.021354	0.035473	-1.023392	
13	imce	~	Afectividad	-0.168728	-0.059848	0.064054	-2.634172	
14	imce	~	Sociabilidad	0.157583	0.100567	0.038349	4.109147	
15	imce	~	educm	-0.008484	-0.041828	0.003072	-2.762015	
16	imce	~	educp	-0.004397	-0.022195	0.00293	-1.500467	
17	imce	~	sexo	0.178982	0.129908	0.016695	10.72065	
18	imce	~	act_fisica_2.0	-0.032036	-0.022827	0.024654	-1.299418	
19	imce	~	act_fisica_3.0	0.001732	0.001232	0.024692	0.070126	
20	imce	~	act_fisica_4.0	-0.068908	-0.022154	0.042095	-1.636986	
21	imce	~	act_fisica_5.0	-0.030300	-0.011244	0.037737	-0.802939	
22	imce	~	madre_work_0	-0.004144	-0.001878	0.028829	-0.143732	
23	imce	~	madre_work_1	-0.017325	-0.012827	0.018587	-0.932137	
24	imce	~	area	0.008442	0.003744	0.027973	0.301773	
25	imce	~	vive_madre	0.073091	0.017416	0.051105	1.430197	
26	imce	~	vive_padre	-0.010788	-0.007221	0.018493	-0.583355	
27	Afectividad	~~	Afectividad	0.057324	1.000000	0.002364	24.244354	
28	Curiosidad	~~	Curiosidad	0.157654	1.000000	0.006585	23.940619	
29	Curiosidad	~~	Afectividad	0.054494	0.573225	0.002155	25.288599	
30	Curiosidad	~~	Sociabilidad	0.099594	0.582274	0.003805	26.174279	
31	Sociabilidad	~~	Sociabilidad	0.185570	1.000000	0.006915	26.837378	
32	Sociabilidad	~~	Afectividad	0.064962	0.629849	0.002362	27.497687	
33	imce	~~	imce	0.443263	0.972851	0.007732	57.327029	
34	sk1	~~	sk1	0.095819	0.625683	0.002015	47.542849	
35	sk10	~~	sk10	0.626305	0.684635	0.012547	49.917485	
36	sk11	~~	sk11	0.210646	0.467350	0.005496	38.330289	
37	sk12	~~	sk12	0.468204	0.694723	0.009311	50.284015	
38	sk13	~~	sk13	0.742171	0.721678	0.014493	51.209617	
39	sk2	~~	sk2	0.244877	0.567435	0.005481	44.678203	

40	sk3	~~	sk3	0.214547	0.631261	0.004489	47.788679
41	sk4	~~	sk4	0.158955	0.506956	0.003871	41.063477
42	sk5	~~	sk5	0.165405	0.471274	0.00522	31.685072
43	sk6	~~	sk6	0.396837	0.689604	0.008391	47.291286
44	sk8	~~	sk8	0.523623	0.697126	0.010983	47.676349
45	sk9	~~	sk9	0.272405	0.633413	0.005691	47.862941

	p-value
0	-
1	0.0
2	0.0
3	0.0
4	0.0
5	-
6	0.0
7	0.0
8	0.0
9	-
10	0.0
11	0.0
12	0.306123
13	0.008434
14	0.00004
15	0.005745
16	0.133493
17	0.0
18	0.193801
19	0.944093
20	0.101633
21	0.42201
22	0.885712
23	0.351266
24	0.762825
25	0.15266
26	0.559654
27	0.0
28	0.0
29	0.0
30	0.0
31	0.0
32	0.0
33	0.0
34	0.0
35	0.0
36	0.0
37	0.0
38	0.0

```
39       0.0
40       0.0
41       0.0
42       0.0
43       0.0
44       0.0
45       0.0

       DoF  DoF Baseline         chi2  chi2 p-value  chi2 Baseline       CFI  \
Value  282           312  2619.890536           0.0   30313.592702  0.922074

            GFI      AGFI       NFI       TLI     RMSEA        AIC  \
Value  0.913574  0.904379  0.913574  0.913785  0.035284  85.213246

              BIC    LogLik
Value  377.779861  0.393377
```

0.7.3 Cluster 3: Desnutrición

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.929
Number of iterations: 55
Params: 1.046 1.113 0.825 1.073 2.467 2.203 2.775 1.087 1.313 -0.046 1.027
-0.429 0.023 -0.004 -0.096 0.216 0.310 0.242 0.426 -0.065 0.024 0.162 0.403
-0.016 0.028 0.225 0.047 0.093 0.151 0.048 0.454 0.081 0.656 0.216 0.349 0.762
0.241 0.198 0.139 0.166 0.409 0.533 0.279

            lval  op            rval  Estimate  Est. Std  Std. Err    z-value  \
0            sk9   ~      Curiosidad  1.000000  0.667622         -          -
1           sk10   ~      Curiosidad  1.046243  0.522097  0.119713   8.739616
2           sk11   ~      Curiosidad  1.113308  0.750703  0.098941  11.252255
3           sk12   ~      Curiosidad  0.825146  0.551732  0.090153   9.152704
4           sk13   ~      Curiosidad  1.072854  0.503252  0.126669    8.46975
5            sk1   ~     Afectividad  1.000000  0.505812         -          -
6            sk2   ~     Afectividad  2.466740  0.642016  0.289199   8.529562
7            sk3   ~     Afectividad  2.202539  0.636195  0.259422   8.490187
8            sk4   ~     Afectividad  2.775424  0.779008  0.301314   9.211066
9            sk5   ~    Sociabilidad  1.000000  0.691299         -          -
10           sk6   ~    Sociabilidad  1.087025  0.551614  0.129293   8.407462
11           sk8   ~    Sociabilidad  1.313067  0.573493  0.152138   8.630749
12          imce   ~      Curiosidad -0.045902 -0.030953  0.115044  -0.398994
13          imce   ~     Afectividad  1.026869  0.243470  0.516422   1.988428
14          imce   ~    Sociabilidad -0.429180 -0.237663  0.212246  -2.022089
15          imce   ~           educm  0.023278  0.107360  0.013231    1.75928
```

16	imce	~	educp	-0.004272	-0.020109	0.012742	-0.335272
17	imce	~	sexo	-0.095890	-0.066275	0.070138	-1.36716
18	imce	~	act_fisica_2.0	0.215799	0.147366	0.101635	2.123267
19	imce	~	act_fisica_3.0	0.310497	0.212342	0.101782	3.050599
20	imce	~	act_fisica_4.0	0.242157	0.072095	0.178386	1.357486
21	imce	~	act_fisica_5.0	0.425572	0.156975	0.151439	2.810176
22	imce	~	madre_work_0	-0.064853	-0.028150	0.119733	-0.541651
23	imce	~	madre_work_1	0.024008	0.017076	0.07526	0.319007
24	imce	~	area	0.161522	0.052640	0.151084	1.069082
25	imce	~	vive_madre	0.403276	0.078984	0.247432	1.629846
26	imce	~	vive_padre	-0.015719	-0.009935	0.077144	-0.203765
27	Afectividad	~~	Afectividad	0.027771	1.000000	0.00561	4.950424
28	Curiosidad	~~	Curiosidad	0.224631	1.000000	0.033271	6.751624
29	Curiosidad	~~	Afectividad	0.047327	0.599202	0.007574	6.24884
30	Curiosidad	~~	Sociabilidad	0.092617	0.502068	0.015395	6.015953
31	Sociabilidad	~~	Sociabilidad	0.151489	1.000000	0.023484	6.450808
32	Sociabilidad	~~	Afectividad	0.047629	0.734310	0.007089	6.718266
33	imce	~~	imce	0.453583	0.918164	0.033227	13.650965
34	sk1	~~	sk1	0.080776	0.744154	0.006131	13.174553
35	sk10	~~	sk10	0.656169	0.727415	0.050903	12.890472
36	sk11	~~	sk11	0.215623	0.436446	0.023551	9.155735
37	sk12	~~	sk12	0.349486	0.695591	0.027665	12.632896
38	sk13	~~	sk13	0.762341	0.746738	0.058483	13.03531
39	sk2	~~	sk2	0.240987	0.587816	0.02032	11.85989
40	sk3	~~	sk3	0.198138	0.595256	0.016595	11.939776
41	sk4	~~	sk4	0.138588	0.393146	0.015734	8.808203
42	sk5	~~	sk5	0.165504	0.522106	0.018185	9.100967
43	sk6	~~	sk6	0.409284	0.695722	0.034316	11.92685
44	sk8	~~	sk8	0.532955	0.671106	0.045882	11.615891
45	sk9	~~	sk9	0.279344	0.554281	0.025177	11.095275

	p-value
0	-
1	0.0
2	0.0
3	0.0
4	0.0
5	-
6	0.0
7	0.0
8	0.0
9	-
10	0.0
11	0.0
12	0.689898
13	0.046764
14	0.043167

```
15   0.07853
16   0.73742
17  0.171575
18  0.033731
19  0.002284
20  0.174627
21  0.004951
22  0.588059
23  0.749722
24  0.285033
25  0.103134
26  0.838537
27  0.000001
28       0.0
29       0.0
30       0.0
31       0.0
32       0.0
33       0.0
34       0.0
35       0.0
36       0.0
37       0.0
38       0.0
39       0.0
40       0.0
41       0.0
42       0.0
43       0.0
44       0.0
45       0.0

       DoF  DoF Baseline        chi2  chi2 p-value  chi2 Baseline       CFI  \
Value  282           312  384.399098      0.000047     2225.52253  0.946487

            GFI      AGFI       NFI       TLI     RMSEA     AIC         BIC  \
Value  0.827277  0.808902  0.827277  0.940794  0.029652  84.143  257.255236

       LogLik
Value  0.9285
```

R: Se puede ver que la estimación de los parámetros cambia aplicando el modelo por clusters, esto debido a que hay algunas variables que pueden aportar de distinta manera según cada cluster, por ejemplo:

```
Lo explicado en el Cluster 2 de la variable Afectividad en el IMCE aporta de manera negativa, m
```

Además, los modelos presentaron buenos resultados en los estadísticos por lo que podemos decir que es un buen ajuste, como se muestra a continuación,

```
1. Cluster 1:
    TLI : 0.9175
    RMSEA : 0.0347
    CFI : 0.9255

2. Cluster 2:
    TLI : 0.9138
    RMSEA : 0.0353
    CFI : 0.9221

3. Cluster 3:
    TLI : 0.9408
    RMSEA : 0.0297
    CFI : 0.9465
```

FInalmente, si nosotros queremos ver una forma en general, es mucho mejor aplicar el modelo a DF original, con todos los clusters, pero si se quiere ver de forma más profunda, es bueno aplicar el modelo para cada Cluster ya que nos permite ver como afectan ciertos factores o variables a cada Cluster en específico y como pueden caracterizar cada cluster, por lo que agrega más valor al análisis.